Systems Analysis and Synthesis, 3 credits

1 Course information:

This course presents an introduction to state space analysis of linear systems. This includes stability, controllability, observability, realization theory, state feedback, the Kalman decomposition and the linear quadratic regular problem.

2 Course Description:

- Converting linear differential equations to state space.
- Computing the exponential matrix e^{At}.
- The Cayley-Hamilton Theorem and its applications.
- Stable systems and Lyapunov stability.
- Controllability and observability.
- Realization theory.
- Minimal, controllable and observable realizations.
- The Kalman decomposition.
- Reachable and detectable systems.
- State feedback and pole placement.
- Observers, state estimators and detectability
- Output feedback controllers.
- The linear quadratic regulator.
- Discrete time systems.

3 Learning Resources and Reference Text

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook. The textbook is a set of notes. The notes are posted on Brightspace in a .pdf file free of charge.

4 Assignments

The course assigns homework which is graded and included in part of the final grade. The homework will be handed in by gradescope or brightspace. The solutions for the homework will be provided. No late homework will be accepted unless there is a medical or university related reason.

5 Course Learning Objectives

On completing this course the student shall be able to:

1. Set up and analyze systems in state space form.
2. Use Lyapunov functions for stability analysis.
3. Use state space to stabilize unstable controllable systems.
4. Design state feedback controllers.
5. Design Observers.
6. Design Output feedback controllers.
7. To have the background and prerequisites for a course on Kalman filtering.

6 Grading Scale

- Grades are A, B, C, D, F and no $\pm$ grades.
- The homework counts for 35% of the final grade. The exams and final exam will account for 65% of the final grade.
- The final grade is determined by a curve. The instructor sorts all the grades and then calculates the mean, median and standard deviation. Based on this data an A is a minimum of 35% of the class (and could be a higher percentage). The rest of the grades are B with the exception of usually a very few students which did very little for the course. They may receive a C, D or F grade.
- The grading and distribution of grades in this course are done totally blind. The instructor has no idea what student is associated with any score when assigning grades.